

# **A Reproducible Machine-Learning Benchmark for Medullary Thyroid Carcinoma Status in Published MEN2/RET-Carrier Cohorts**

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# Abstract

Multiple Endocrine Neoplasia type 2 (MEN2) is a rare hereditary endocrine cancer syndrome in which published evidence is often distributed across small cohorts, pedigrees, and case reports. This study developed an exploratory, reproducible machine-learning benchmark for medullary thyroid carcinoma (MTC) status among published MEN2/RET-carrier records. We curated 149 RET carriers from 10 peer-reviewed sources spanning 14 RET variants, harmonized demographic, genotype, clinical, and biomarker variables, and evaluated logistic regression, random forest, XGBoost, LightGBM, and linear support vector machine models under an 80/20 stratified internal held-out split. The primary genotype-aware XGBoost analysis on the literature-derived cohort correctly identified 15 of 15 MTC cases in the internal test split, with 83.3% accuracy (25/30), 100.0% sensitivity (15/15; 95% Wilson interval 79.6%-100.0%), and 66.7% specificity (10/15; 95% Wilson interval 41.7%-84.8%). A genotype-blind sensitivity analysis that removed RET variant and ATA risk features retained 93.3% sensitivity (14/15) but lower accuracy (73.3%). A secondary synthetic-augmentation experiment reached 96.2% accuracy with LightGBM but is interpreted only as a simulation because synthetic controls may not represent real clinical heterogeneity. The main contribution is therefore not a deployable diagnostic tool, but a transparent rare-disease benchmarking workflow that exposes feature-dependence, imputation uncertainty, and biomarker-timing limitations. Independent prospective validation is required before clinical or screening use.

## Keywords

MEN2; RET; Medullary thyroid carcinoma; Calcitonin; Carcinoembryonic antigen; Rare disease; Machine learning; Screening; Risk stratification; Reproducibility

## 1 Introduction

Multiple Endocrine Neoplasia type 2 (MEN2) is a hereditary endocrine cancer syndrome caused by germline pathogenic variants in the RET proto-oncogene. MEN2 is clinically associated with medullary thyroid carcinoma (MTC), and may also involve pheochromocytoma and primary hyperparathyroidism depending on the MEN2 subtype and RET variant. Current clinical management is guided by genotype-phenotype correlations, serum calcitonin and carcinoembryonic antigen (CEA), family screening, and guideline-based timing of thyroidectomy or surveillance (1).

Computational modeling in MEN2 is limited by the rarity of the syndrome. The practical motivation is also human: Can we save those 20k Rs people with just a simple blood test? In India, MEN2-related RET genetic testing is advertised at about INR 20,000, or approximately 225 USD, which can put life-saving diagnosis out of reach for many families (19). This research therefore asks: *can machine learning on routine blood biomarkers, including calcitonin and CEA, and clinical features predict MTC risk without expensive genetic sequencing?* The available literature is scientifically valuable for asking this question, but it creates methodological risks for machine

learning: small sample size, overlapping familial patterns, inconsistent biomarker reporting, post-diagnostic measurements, missing CEA values, and possible circularity when genotype-derived risk categories are used to predict cancer status.

This study asks a narrower question than clinical diagnosis: can published MEN2/RET-carrier data be converted into a reproducible benchmark that quantifies how model performance changes when genotype, biomarker, and synthetic-augmentation assumptions are varied? In plain terms, the project asks whether routine blood and clinical signals contain useful structure in a rare cancer dataset, while making clear that they cannot yet be used as a low-cost substitute for RET sequencing, clinical evaluation, or guideline-based care.

To address the main validity concern, the analysis separates two modeling tracks. The genotype-aware benchmark includes RET variant and ATA-aligned risk features, and is relevant only after RET carrier status is already known. The genotype-blind sensitivity analysis removes RET variant and ATA risk features to test whether non-genetic clinical and biomarker variables carry signal. A secondary synthetic-augmentation experiment is reported separately as a simulation, not as an expanded real-world cohort.

## 2 Materials and Methods

### 2.1 Study Design

This was a retrospective secondary analysis of published, de-identified MEN2/RET-carrier records. No new patients were recruited, and no new biological specimens were collected. The primary dataset was the literature-derived cohort of 149 confirmed RET carriers. The target variable was binary MTC status: MTC present vs. MTC absent in the extracted source record.

### 2.2 Data Sources and Eligibility

Studies were eligible if they reported patient-level or reconstructable carrier-level information on RET-positive individuals, MTC status, age or demographic variables, and at least one clinical, genotype, or biomarker feature usable for harmonization. Sources were excluded if they lacked RET-carrier-level information, duplicated already extracted records, reported only aggregate non-reconstructable summaries, or focused on non-RET thyroid cancer without MEN2-carrier information.

Table 1: Published sources used to construct the literature-derived cohort.

| Study | Source                      | Main variant context                      | n  | DOI                                                                       |
|-------|-----------------------------|-------------------------------------------|----|---------------------------------------------------------------------------|
| 1     | Xu et al., 2016 (3)         | RET K666N families                        | 24 | <a href="https://doi.org/10.1089/thy.2016.0374">10.1089/thy.2016.0374</a> |
| 2     | Frank-Raue et al., 2006 (2) | Multiple RET variants after thyroidectomy | 46 | <a href="https://doi.org/10.1530/eje.1.02216">10.1530/eje.1.02216</a>     |

| Study | Source                    | Main variant context          | n  | DOI                                                                                 |
|-------|---------------------------|-------------------------------|----|-------------------------------------------------------------------------------------|
| 3     | Qi et al., 2015 (5)       | RET S891A/R525W familial MTC  | 15 | <a href="https://doi.org/10.18632/oncotarget.4992">10.18632/oncotarget.4992</a>     |
| 4     | Florescu et al., 2024 (8) | RET C634G kindred             | 6  | <a href="https://doi.org/10.3390/clinpract14060179">10.3390/clinpract14060179</a>   |
| 5     | La Greca et al., 2024 (9) | RET K666N family              | 4  | <a href="https://doi.org/10.1530/EDM-24-0009">10.1530/EDM-24-0009</a>               |
| 6     | Vijayan et al., 2021 (7)  | Indian RET S891A pedigree     | 7  | <a href="https://doi.org/10.4103/ijc.IJC_639_19">10.4103/ijc.IJC_639_19</a>         |
| 7     | Zhang et al., 2024 (10)   | RET C634Y family              | 3  | <a href="https://doi.org/10.12998/wjcc.v12.i15.2627">10.12998/wjcc.v12.i15.2627</a> |
| 8     | Shankar et al., 2012 (11) | Familial MEN2B infant         | 2  | <a href="https://doi.org/10.1186/1687-9856-2012-25">10.1186/1687-9856-2012-25</a>   |
| 9     | Schulte et al., 2010 (4)  | RET S891A multi-center cohort | 36 | <a href="https://doi.org/10.1210/jc.2010-0375">10.1210/jc.2010-0375</a>             |
| 10    | Qi et al., 2014 (6)       | Chinese Han RET S891A family  | 6  | <a href="https://doi.org/10.1007/s12038-014-9428-x">10.1007/s12038-014-9428-x</a>   |

## 2.3 Data Extraction and Harmonization

Each source was converted into structured JSON and then into a tabular dataset. Extracted variables included age, sex, RET variant, ATA-aligned risk level, calcitonin elevation and numeric value, CEA numeric value, CEA missingness flag, thyroid nodule indicators, family history of MTC, pheochromocytoma, hyperparathyroidism, MEN2 syndrome status, C-cell disease status, and MTC diagnosis. RET variants were normalized to protein-level short notation where possible.

The primary genotype-aware feature set included age, sex, RET variant one-hot encodings, ATA-aligned risk level, calcitonin features, CEA features, thyroid nodule variables, family history of MTC, pheochromocytoma, hyperparathyroidism, age group, and simple interaction terms. The genotype-blind sensitivity analysis removed RET variant encodings, ATA risk level, and ATA-risk interaction terms. The no-CEA sensitivity analysis removed CEA numeric values. The biomarker-timing sensitivity analysis removed calcitonin, CEA, and nodule-derived features, leaving genotype and demographic features.

## 2.4 Missing Data and CEA Imputation

CEA was observed in only 12 of 149 records before imputation. MICE with predictive mean matching was used as one exploratory imputation strategy and compared with no-CEA and simpler imputation alternatives in sensitivity analyses (12). CEA imputation was not treated as independently validated because the observed paired calcitonin-CEA subset was small and weakly correlated ( $r = 0.1158$ ,  $n = 12$ ). A binary CEA-imputed flag was retained to preserve missingness information.

Table 2: Key variable groups used in the benchmark.

| Variable group        | Variables                                                                                        |
|-----------------------|--------------------------------------------------------------------------------------------------|
| Demographic           | Age, sex, age group, age squared                                                                 |
| Genotype-aware        | RET variant one-hot encodings, ATA-aligned risk level, risk-age and risk-calcitonin interactions |
| Biomarker             | Calcitonin elevated flag, numeric calcitonin, CEA numeric value, CEA missingness/imputation flag |
| Clinical presentation | Thyroid nodule indicators, family history of MTC, pheochromocytoma, hyperparathyroidism          |
| Outcome               | Binary MTC status extracted from source reports                                                  |

Table 3: Selected missingness and harmonization summary in the 149-record literature-derived cohort.

| Field                          | Observed or nonzero | Total | Note                                                                           |
|--------------------------------|---------------------|-------|--------------------------------------------------------------------------------|
| Age                            | 149                 | 149   | Missing ages were filled from source-level medians where required.             |
| Sex                            | 149                 | 149   | Unknown values were mode-filled for modeling.                                  |
| RET variant                    | 149                 | 149   | Required for genotype-aware inclusion.                                         |
| Calcitonin numeric value       | 77                  | 149   | Zero includes unavailable, normal, or undetectable values after harmonization. |
| CEA observed before imputation | 12                  | 149   | Observed paired CEA values seeded the imputation analysis.                     |
| CEA imputed                    | 137                 | 149   | Missingness flag retained in the model dataset.                                |

## 2.5 Biomarker Timing and Leakage Handling

Biomarker timing varied across sources. Some studies reported preoperative calcitonin, some reported surveillance values from cascade testing, and some case reports mixed diagnostic, preoperative, postoperative, or follow-up measurements. Because patient-level timing was not uniformly recoverable across all extracted records, the primary manuscript reports biomarker timing as a limitation and includes a conservative biomarker-exclusion sensitivity analysis.

## 2.6 Models and Statistical Approach

Five model families were benchmarked: logistic regression, random forest, XGBoost, LightGBM, and linear support vector machine. Implementations used scikit-learn, XGBoost, and LightGBM (14-16). The dataset was split into 80% training and 20% internal held-out testing using stratification

Table 4: Source-level biomarker timing classification used for leakage assessment.

| Study | Timing class          | Leakage relevance                                                                                                |
|-------|-----------------------|------------------------------------------------------------------------------------------------------------------|
| 1     | Mixed                 | Probands often had diagnostic/preoperative or follow-up values; relatives had cascade-surveillance values.       |
| 2     | Preoperative          | Basal/stimulated calcitonin before thyroidectomy; CEA not reported.                                              |
| 3     | Preop./surv.          | Family screening and surgery workup values; CEA mostly not individually reported.                                |
| 4     | Surv./preop.          | Kindred-level MEN2A evaluation; CEA not central.                                                                 |
| 5     | Preop./surv.          | Family case-report values around diagnosis and carrier follow-up.                                                |
| 6     | Surv./preop.          | Pedigree screening values around family evaluation.                                                              |
| 7     | Post-diagnostic/Mixed | Small family report with values around established MEN2A diagnosis and surgery.                                  |
| 8     | Mixed                 | Infant preoperative calcitonin plus postoperative/surveillance follow-up; mother post-diagnostic values.         |
| 9     | Preoperative          | Multicenter S891A cohort reports preoperative calcitonin multiples; CEA measured in subset.                      |
| 10    | Preop./postop.        | Pedigree table includes pre- and postoperative calcitonin; extraction favored preoperative values where present. |

and fixed random seed 42. Feature scaling was fit on training data only and then applied to the held-out split. SMOTE was applied only to training folds where class balancing was used (13). Performance metrics included accuracy, precision, sensitivity, specificity, F1-score, ROC-AUC, and average precision. Raw confusion-matrix counts were reported with Wilson 95% confidence intervals for binomial proportions so that small held-out sample sizes were visible.

The primary endpoint was internal held-out sensitivity on the literature-derived cohort. Secondary endpoints were accuracy, specificity, ROC-AUC, genotype-blind performance, no-CEA performance, biomarker-exclusion performance, and performance in the synthetic-augmentation simulation. SHAP was used only as an exploratory interpretation method for supported models (17).

## 2.7 Synthetic-Augmentation Experiment

A secondary case-control simulation expanded the data to 1,047 records using variant-matched synthetic controls and synthetic MTC-like examples generated from source distributions, followed by SMOTE only within training data. These records are not real patients. Results from this experiment are therefore interpreted as a stress test of model behavior under augmentation, not as evidence of external clinical validity.

## 3 Results

### 3.1 Cohort Summary

The literature-derived cohort contained 149 RET carriers from 10 studies. There were 73 records with MTC and 76 without MTC. The cohort included 99 female and 50 male carriers, with mean age 34.27 years, median age 31.5 years, and range 0.17 to 90 years. Fourteen RET variants were represented: S891A, K666N, C634Y, C634R, R525W, C634G, C630R, C634W, L790F, V804M, Y791F, C618S, C620Y, and M918T.

Figure 1 shows the RET variant distribution, Figure 2 shows the age distribution, and Figure 3 shows the limited paired calcitonin-CEA data used for exploratory CEA imputation.

### 3.2 Primary Genotype-Aware Benchmark

In the primary internal held-out split of the 149-record literature-derived cohort, the genotype-aware XGBoost model produced a confusion matrix of TN = 10, FP = 5, FN = 0, and TP = 15. This corresponds to 83.3% accuracy (25/30; 95% Wilson interval 66.4%-92.7%), 100.0% sensitivity (15/15; 95% Wilson interval 79.6%-100.0%), and 66.7% specificity (10/15; 95% Wilson interval 41.7%-84.8%). The point estimate therefore had no false negatives in this one split, but the confidence interval remains wide because only 15 positive test cases were present.

Table 5: Internal held-out performance with raw counts and Wilson confidence intervals.

| Analysis       | Model    | Confusion matrix          | Sensitivity                  | Accuracy                     |
|----------------|----------|---------------------------|------------------------------|------------------------------|
| Genotype-aware | XGBoost  | TN=10, FP=5, FN=0, TP=15  | 100.0% (15/15; 79.6%-100.0%) | 83.3% (25/30; 66.4%-92.7%)   |
| Genotype-blind | XGBoost  | TN=8, FP=7, FN=1, TP=14   | 93.3% (14/15; 70.2%-98.8%)   | 73.3% (22/30; 55.6%-85.8%)   |
| No CEA         | XGBoost  | TN=12, FP=3, FN=0, TP=15  | 100.0% (15/15; 79.6%-100.0%) | 90.0% (27/30; 74.4%-96.5%)   |
| No biomarkers  | XGBoost  | TN=12, FP=3, FN=0, TP=15  | 100.0% (15/15; 79.6%-100.0%) | 90.0% (27/30; 74.4%-96.5%)   |
| Synthetic      | LightGBM | TN=156, FP=3, FN=5, TP=46 | 90.2% (46/51; 79.0%-95.7%)   | 96.2% (202/210; 92.7%-98.1%) |

### 3.3 Genotype-Blind and Feature Sensitivity Analyses

Removing RET variant encodings, ATA-aligned risk level, and risk interaction terms reduced XGBoost performance on the original held-out split to 73.3% accuracy (22/30) and 93.3% sensitivity (14/15). This result shows that non-genetic clinical and biomarker features carry some signal, but it also confirms that the genotype-aware model benefits from genetic context. Therefore, genotype-aware performance should not be presented as a substitute for genetic testing.

Removing CEA improved the XGBoost result in the original split to 90.0% accuracy while preserving 15/15 sensitivity. This supports treating CEA imputation as exploratory rather than essential to the main result. Removing calcitonin, CEA, and nodule-derived features as a conservative biomarker-timing sensitivity analysis also preserved 15/15 sensitivity in this split, but this should not be interpreted as evidence that biomarkers are clinically unnecessary. It only shows that this small internal test split was not driven entirely by biomarker timing.

### 3.4 Synthetic-Augmentation Simulation

In the synthetic-augmentation experiment, LightGBM achieved 96.2% accuracy (202/210) and 90.2% sensitivity (46/51) on the internal held-out split. These values were higher than the corresponding LightGBM result on the original 149-record cohort, but the augmented dataset contains synthetic records and synthetic controls. The result is therefore useful for studying how augmentation changes discrimination, not for estimating real-world clinical performance.

### 3.5 Interpretability

Exploratory SHAP analysis was generated for supported models. In the synthetic LightGBM analysis, genotype and biomarker-related features both contributed to predictions. Because the synthetic analysis may reflect assumptions used to generate controls, SHAP rankings from that experiment should not be treated as clinical feature rankings.

## 4 Discussion

This study presents a reproducible benchmark for machine-learning analysis of MTC status in published MEN2/RET-carrier records. The strongest point estimate in the original cohort came from a genotype-aware XGBoost model, which identified all 15 MTC cases in one internal held-out split. However, the Wilson interval for sensitivity was 79.6%-100.0%, making the sample-size limitation explicit. The result should be read as an exploratory internal benchmark, not as evidence that the model is ready for clinical screening.

The most important methodological issue is the role of genotype. RET variant and ATA risk level are clinically meaningful variables, but they can also encode prior knowledge about disease risk. The genotype-aware model is therefore relevant only for confirmed or suspected RET-carrier datasets where genetic context is already known. The genotype-blind sensitivity analysis partially addresses this concern by removing RET/ATA features. Its lower accuracy and one false negative show that the full model does use genetic context and should not be described as a low-cost replacement for RET sequencing.

CEA imputation is another major limitation. Only 12 observed paired CEA values were available, and the observed calcitonin-CEA correlation was weak. Although MICE with predictive mean matching is a reasonable exploratory imputation method, the no-CEA sensitivity analysis is more important for credibility than the imputed CEA result. Future versions should either collect more observed CEA values or treat CEA as optional/missing rather than central.

Biomarker timing also limits interpretation. Some calcitonin and CEA values were preoperative or surveillance measurements, while others were post-diagnostic or mixed with follow-up information. If a biomarker was measured after cancer diagnosis or surgery, it could leak outcome information into the model. Because timing could not be standardized across all records, the manuscript reports a conservative biomarker-exclusion sensitivity analysis and treats all biomarker-based findings as provisional.

The synthetic-augmentation experiment should be interpreted cautiously. Synthetic controls can make model training easier and improve apparent discrimination, but they may also encode



assumptions about what non-MTC carriers look like. For this reason, the augmented LightGBM result is presented as a simulation of augmentation behavior, not as an expanded clinical cohort.

Several biological observations remain relevant for future work. RET K666N can show heterogeneous penetrance, and allele dosage may matter in some reports (3,9). RET S891A has been reported in Indian and Chinese pedigrees and may show variable clinical behavior across families (5-7). Codon 634 variants are associated with MEN2A phenotypes and have been reported with cutaneous lichen amyloidosis in some kindreds (8,10). These examples show why simple binary variant encoding is likely insufficient for future clinical modeling.

## **4.1 Limitations**

The main limitations are small sample size, retrospective literature-derived data, heterogeneous source reporting, possible familial clustering, incomplete biomarker timing, extensive CEA imputation, and lack of independent external validation. The held-out split is internal to the curated literature dataset and does not test generalization to a new institution, new country, or prospective clinical population. The public dataset was reconstructed from published reports; although no new identifiers were collected, rare-disease pedigrees can carry re-identification risk. Public-facing datasets should therefore use anonymized row identifiers and avoid unnecessary family labels or exact relationship details.

## **5 Conclusion**

MEN2 Predictor is best framed as a reproducible exploratory benchmark for rare-disease machine learning, not as a deployable diagnostic or screening system. The workflow shows how performance changes when genotype features, CEA imputation, biomarker-dependent features, and synthetic augmentation are varied. The results support further study of structured MEN2 data aggregation, but independent prospective validation with standardized pre-diagnostic variables is required before any clinical or screening use.

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## **Author Contributions**

H.K. contributed to literature search, study identification, data curation, and biological interpretation. A.V.P. designed and implemented the machine-learning pipeline, trained the models, and performed the computational analysis. S.M. provided mentorship, advised on methodological decisions, and oversaw the project direction. All authors reviewed and approved the final manuscript.

## **Conflict of Interest**

The authors declare no conflicts of interest.

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## **Data Availability**

The analysis code, processed data artifacts, and figure-generation scripts are available in the GitHub repository at <https://github.com/arjuncodess/men2-predictor>. Anonymized submission copies of the processed original and augmented datasets are provided under `paper/submission_data/` to reduce avoidable patient- or family-level identifiability in review materials. The hosted Hugging Face Space is provided for educational demonstration only and must not be used for diagnosis, treatment, triage, or any real-world clinical decision.

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## A Supplementary Materials

Table 6: RET-carrier counts by source study label used in the aggregated cohort (total n=149).

| Study label                                                 | Patients (n) |
|-------------------------------------------------------------|--------------|
| European Journal of Endocrinology (2006)                    | 46           |
| JCEM (2010) RET S891A MEN2A Spectrum                        | 36           |
| Thyroid Journal (2016)                                      | 24           |
| Oncotarget (2015) RET S891A FMTC                            | 15           |
| Indian Journal of Cancer (2021) RET S891A Family            | 7            |
| Clinics and Practice (2024) RET C634G Kindred               | 6            |
| Journal of Biosciences (2014) RET S891A Chinese FMTC Family | 6            |
| Endocrinol. Diabetes Metab. Case Reports (2024) RET K666N   | 4            |
| World Journal of Clinical Cases (2024) RET C634Y Family     | 3            |
| Int. J. Pediatr. Endocrinol. (2012) Familial MEN2B Infant   | 2            |

Table 7: Ablation study results: XGBoost on the original paper-only dataset (n=149). Accuracy change is relative to baseline.

| Configuration                 | Features | Accuracy | Recall | F1    | $\Delta$ Acc |
|-------------------------------|----------|----------|--------|-------|--------------|
| Baseline (All Features)       | 33       | 0.833    | 1.000  | 0.857 | —            |
| Without ATA Risk Level        | 30       | 0.767    | 0.867  | 0.788 | −6.7%        |
| Without Variant Encodings     | 19       | 0.833    | 1.000  | 0.857 | 0.0%         |
| Without Any Genetic Features  | 16       | 0.733    | 0.933  | 0.778 | −10.0%       |
| Without Calcitonin            | 30       | 0.833    | 1.000  | 0.857 | 0.0%         |
| Without CEA                   | 32       | 0.900    | 1.000  | 0.909 | +6.7%        |
| Genetics Only (No Biomarkers) | 27       | 0.900    | 1.000  | 0.909 | +6.7%        |
| Biomarkers Only (No Genetics) | 16       | 0.733    | 0.933  | 0.778 | −10.0%       |

Table 8: Ablation study results: LightGBM on the expanded dataset (n=1,047). Accuracy change is relative to baseline.

| Configuration                 | Features | Accuracy | Recall | F1    | $\Delta$ Acc |
|-------------------------------|----------|----------|--------|-------|--------------|
| Baseline (All Features)       | 34       | 0.962    | 0.902  | 0.920 | —            |
| Without ATA Risk Level        | 31       | 0.952    | 0.882  | 0.900 | −1.0%        |
| Without Variant Encodings     | 20       | 0.943    | 0.863  | 0.880 | −1.9%        |
| Without Any Genetic Features  | 17       | 0.933    | 0.824  | 0.857 | −2.9%        |
| Without Calcitonin            | 31       | 0.952    | 0.882  | 0.900 | −1.0%        |
| Without CEA                   | 33       | 0.929    | 0.882  | 0.857 | −3.3%        |
| Genetics Only (No Biomarkers) | 27       | 0.910    | 0.843  | 0.819 | −5.2%        |
| Biomarkers Only (No Genetics) | 17       | 0.933    | 0.824  | 0.857 | −2.9%        |